

Day-ahead load forecasting for the Hungarian market

What the question became

The client must commit to hourly purchases for tomorrow before noon today. I narrowed the task to forecasting hourly net load for the next day using only information available before the cutoff. The TSO day-ahead estimate was the natural anchor, so I modeled the gap between that estimate and actual load and added the predicted correction back to the public forecast.

What I built

The pipeline reads the 15-minute file, converts it to hourly averages, and keeps volatility measures from the underlying readings. Features include recent load, same-hour history, morning load, calendar and holiday information, the shape of the TSO forecast for the target day, recent TSO errors, and forecasted weather from Open-Meteo. The model is a gradient boosted tree tuned with Optuna using rolling time-based validation, with a final holdout period kept separate from tuning.

What I would tell the client

They should use it if it continues to beat the TSO forecast on recent weeks. In backtests it lowers mean absolute hourly imbalance relative to the public forecast. The practical value is fewer avoidable balancing positions and a more stable day-ahead submission. Before using it with money, they should know that the model needs the TSO forecast for the target day and that the default loss treats overbuying and underbuying equally. The code can produce quantile forecasts once the penalty ratio is known.

What I do not fully trust

I do not fully trust the publication timing of the TSO forecast column. If any of it was filled in after the fact, the residual target may look too optimistic. The national series is also not a perfect proxy for the client portfolio, because industrial customers, and retail mix can shift the error profile. Weather features help, but the historical forecast archive may not exactly match what a desk would have seen at noon on every past day.

What I would check first

I would split errors by hour, weekday, holiday, temperature band, and solar period to see where mistakes concentrate. I would then prune weak features and look for market data such as prices, cross-border flows, and renewable generation forecasts. I would also test a second model family. A neural sequence model could capture longer dependencies, and a simple average or a linear stack of the tree model and neural model may be more stable than either alone. If we later forecast at both 15 minutes and hourly (maybe also daily), I

would add hierarchical reconciliation using a loss that matches the business problem, such as absolute error or quantile loss.

AI tools

I used a large language model assistant for code scaffolding, debugging, and drafting. The model design, feature choices, validation scheme, and interpretation were mainly mine, but I often consulted with LLMs throughout the process. The tools used were mainly github copilot and qwen studio.